

HEOR and Real-World Evidence Analyst Roadmap

Action Kit

Evaluate the clinical and economic value of healthcare interventions using real-world data, cost-effectiveness models, and outcomes research. The role that determines which treatments get adopted, reimbursed, and scaled.

Difficulty	High
Timeline	6 to 12 months
Last Updated	2026-03-25

This action kit contains your 90-day execution plan, portfolio project templates, resume bullet formulas, interview preparation questions, and resource checklist.

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Section 1: 90-Day Execution Plan

This plan maps directly to your learning path phases. Each month has specific deliverables and checkpoints to keep you on track.

Month 1: Foundation

Hours: 80 to 120

- Health economics fundamentals (cost-effectiveness analysis, cost-utility analysis, cost-benefit analysis, budget impact analysis)
- Epidemiology and biostatistics review (study designs, confounding, bias, regression, survival analysis)
- Statistical software proficiency (R or SAS for HEOR; Stata is also common)
- Systematic literature review methodology (PRISMA, PICO framework, risk of bias assessment)
- Introduction to real-world data sources (claims databases, EHR data, registries, patient surveys)
- Health technology assessment (HTA) frameworks (NICE, CADTH, ICER, AMCP dossier format)

Checkpoint: Complete an ISPOR short course on cost-effectiveness analysis or systematic literature review. Conduct a practice systematic literature review on a clinical question from your background (minimum 20 articles screened, 8 to 10 included). Write up findings using PRISMA guidelines.

Month 2: Modeling and RWE Methods

Hours: 120 to 180

- Decision-analytic modeling (decision trees, Markov models, microsimulation)
- Health economic modeling software (TreeAge Pro, Excel-based models with VBA)
- Real-world evidence study design (retrospective cohort, case-control, pragmatic trials)
- Claims database analysis (Medicare, Truven/IBM MarketScan, Optum, CPRD in the UK)
- Meta-analysis and indirect treatment comparison methods (network meta-analysis, MAIC)
- Patient-reported outcomes and health-related quality of life (EQ-5D, SF-36, disease-specific instruments)

Checkpoint: Build a cost-effectiveness model (decision tree or Markov) for a clinical question relevant to your background using TreeAge or Excel. Conduct a retrospective database study using a publicly available claims or EHR dataset. Present both projects in ISPOR poster format.

Month 3: Specialization

Hours: 80 to 120

- Track A: HEOR and Economic Modeling (advanced Markov models, partitioned survival analysis, microsimulation, HTA dossier preparation)
- Track B: Real-World Evidence and Observational Research (propensity score methods, instrumental variables, difference-in-differences, target trial emulation)
- Track C: Market Access and Value Communication (payer engagement strategy, value dossier development, AMCP format, pricing and reimbursement)

- Track D: Patient-Centered Outcomes Research (PRO development and validation, qualitative research methods, patient preference studies, PCORI methodology)

Checkpoint: Submit a poster abstract to an ISPOR conference (regional or annual). Complete a specialization-track portfolio project. Apply to 5 HEOR/RWE analyst positions or fellowship programs targeting your chosen track.

Section 2: Portfolio Project Templates

Each project below includes a dataset, tools, timeline, and your clinical advantage. Complete at least 2 to 3 of these for a competitive portfolio.

Project 1: Systematic Literature Review and Meta-Analysis

Conduct a systematic literature review on a clinical question from your background (e.g., comparative effectiveness of two treatment strategies for a condition you managed clinically). Follow PRISMA guidelines. Screen at least 200 titles/abstracts, include 15 to 25 studies, extract data, assess risk of bias, and perform a quantitative meta-analysis with forest plots.

Dataset	PubMed, Cochrane Library, Embase
URL	https://pubmed.ncbi.nlm.nih.gov/
Tools	PubMed/Ovid for searching, Covidence or Rayyan for screening, R (metafor package) or RevMan for meta-analysis
Time Estimate	4 to 6 weeks

Clinical Advantage: You can write clinically precise search strategies because you know the medical terminology, relevant comparators, and meaningful outcomes. Your clinical judgment helps assess whether included studies reflect real practice patterns.

Project 2: Cost-Effectiveness Model for a Treatment Decision

Build a decision-analytic model (Markov or decision tree) comparing two treatment strategies for a condition you know clinically. Populate it with published clinical and cost data. Calculate incremental cost-effectiveness ratios (ICERs). Run sensitivity analyses (one-way, probabilistic). Present results in a format suitable for an HTA submission.

Dataset	Published clinical trial data plus cost data from CMS fee schedules
URL	https://www.cms.gov/medicare/payment/fee-schedules
Tools	TreeAge Pro or Excel with VBA, R for probabilistic sensitivity analysis, Published efficacy data from clinical trials
Time Estimate	4 to 6 weeks

Clinical Advantage: You understand the clinical pathway your model represents. You know which health states patients actually transition through, what complications look like in practice, and which assumptions are clinically reasonable versus artificially simplistic.

Project 3: Real-World Evidence Study Using Claims Data

Design and execute a retrospective cohort study using a publicly available claims or clinical dataset. Define a clear research question, identify the cohort, apply inclusion/exclusion criteria, handle confounding (propensity score matching or regression adjustment), and analyze treatment outcomes. Report using STROBE or RECORD guidelines.

Dataset	CMS Synthetic Medicare Claims Data or MIMIC-IV
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URL	https://data.cms.gov/collection/synthetic-medicare-enrollment-fee-for-service-claims-and-prescription-drug
Tools	SAS, R, or Python for data analysis, SQL for data extraction, STROBE/RECORD reporting guidelines
Time Estimate	4 to 6 weeks

Clinical Advantage: You can design clinically valid inclusion criteria because you understand which diagnosis codes actually capture the condition of interest versus adjacent conditions. You know which confounders matter clinically, not just statistically.

Project 4: Budget Impact Analysis for a Health System Formulary Decision

Develop a budget impact model for adding a new therapy to a health system formulary. Estimate the eligible population, market share uptake, per-patient treatment costs, offset costs (reduced hospitalizations, ER visits), and net budget impact over a 3 to 5 year horizon. Present in AMCP dossier format.

Dataset	Published clinical trial data plus CMS cost data
URL	https://www.cms.gov/data-research
Tools	Excel with structured worksheets, CMS fee schedules for cost inputs, AMCP Format template
Time Estimate	3 to 4 weeks

Clinical Advantage: Your formulary experience means you understand the real-world dynamics of drug adoption: how quickly prescribers switch, which patient segments adopt first, and what the actual offset savings look like versus the theoretical ones.

Project 5: ISPOR-Format Research Poster Presentation

Convert one of the above projects into an ISPOR-format research poster. Follow ISPOR poster guidelines for structure (objective, methods, results, conclusions, limitations). Design a professional poster suitable for conference submission. Write a 300-word abstract in ISPOR format.

Tools	PowerPoint or specialized poster software, ISPOR abstract submission guidelines, Data visualization tools
Time Estimate	1 to 2 weeks

Clinical Advantage: You can frame your research question in clinically meaningful terms that resonate with the mixed audience at ISPOR conferences (clinicians, economists, payers, industry scientists).

Section 3: Resume Bullet Formulas

Use these formulas to translate your clinical experience into language that resonates with hiring managers in health data roles. Replace bracketed text with your specifics.

Nursing Background

- Leveraged [clinical skill: Patient outcomes tracking and quality measure reporting (HCAHPS, Core Measures, readmission rates)] to deliver [tech outcome: Outcomes measurement methodology and endpoint selection], resulting in [quantified impact].
- Leveraged [clinical skill: Evidence-based practice implementation and clinical guideline application] to deliver [tech outcome: Systematic literature review and evidence synthesis], resulting in [quantified impact].
- Leveraged [clinical skill: Care coordination across transitions and resource utilization awareness] to deliver [tech outcome: Healthcare resource utilization analysis and cost modeling], resulting in [quantified impact].
- Leveraged [clinical skill: Patient education and shared decision-making conversations] to deliver [tech outcome: Patient-reported outcome (PRO) instrument selection and interpretation], resulting in [quantified impact].

Pharmacy Background

- Leveraged [clinical skill: Pharmacoeconomics training in PharmD curriculum] to deliver [tech outcome: Cost-effectiveness analysis and budget impact modeling foundations], resulting in [quantified impact].
- Leveraged [clinical skill: Drug information evaluation and literature appraisal (Drug Information rotations)] to deliver [tech outcome: Systematic literature review and critical appraisal methodology], resulting in [quantified impact].
- Leveraged [clinical skill: Formulary management and P&T; committee presentations] to deliver [tech outcome: Health technology assessment (HTA) dossier development and value communication], resulting in [quantified impact].
- Leveraged [clinical skill: Medication therapy management and clinical outcomes documentation] to deliver [tech outcome: Real-world outcomes data collection and longitudinal patient analysis], resulting in [quantified impact].

Other Clinical Background

- Leveraged [clinical skill: Clinical trial coordination and protocol adherence monitoring] to deliver [tech outcome: Study protocol development and real-world study design], resulting in [quantified impact].
- Leveraged [clinical skill: Diagnostic testing ordering and interpretation of clinical significance] to deliver [tech outcome: Clinical endpoint definition and outcome adjudication], resulting in [quantified impact].
- Leveraged [clinical skill: Utilization review and prior authorization decision-making] to deliver [tech outcome: Payer evidence requirements and market access strategy], resulting in [quantified impact].
- Leveraged [clinical skill: Population health management and disease registry maintenance] to deliver [tech outcome: Cohort identification and longitudinal outcomes analysis], resulting in [quantified impact].

Pro Tip: Always quantify. Instead of 'analyzed data,' write 'analyzed 50,000+ claims records to identify \$2.3M in recoverable overpayments.'

Section 4: Interview Preparation Questions

Practice these role-specific questions. For each, prepare a 2-minute answer that highlights both your technical skills and clinical background.

Technical Questions

1. Walk me through how you would design a cost-effectiveness analysis comparing two treatment strategies for a chronic condition. What model structure would you choose and why?
2. Describe the key methodological differences between a network meta-analysis and a standard pairwise meta-analysis. When would you use each?
3. You are designing a retrospective cohort study using claims data to evaluate real-world treatment outcomes. How do you handle confounding, and what are the limitations of propensity score matching in this context?
4. Explain the concept of an incremental cost-effectiveness ratio (ICER) and how you would interpret results on a cost-effectiveness plane for a decision-maker.
5. How would you select and validate a patient-reported outcome measure for a real-world evidence study in a therapeutic area you have clinical experience in?

Behavioral Questions

1. Tell me about a time your clinical experience helped you identify a flaw in a study design or model assumption that a non-clinical researcher might have missed.
2. How do you handle a situation where the economic evidence supports a different conclusion than the clinical evidence? How do you communicate this to stakeholders?
3. Describe how you would present cost-effectiveness findings to a payer audience that is skeptical of industry-sponsored research.
4. Tell me about a time you had to learn a new analytical method or software tool under a tight project deadline. How did you approach it?

Scenario Questions

1. A pharmaceutical client asks you to conduct a cost-effectiveness analysis for a new oncology drug. During your analysis, you find the drug is not cost-effective at the proposed price using standard willingness-to-pay thresholds. How do you present this finding to the client?
2. You are conducting a real-world evidence study and discover that the claims database you are using has a systematic coding pattern that could bias your results (e.g., a diagnosis code is being used inconsistently across sites). How do you address this?
3. An HTA body has requested additional evidence for a product you are supporting. They want a budget impact analysis within 6 weeks. Your team has limited resources. How do you scope the project and deliver?

Section 5: Resource Checklist

Use this checklist to track your progress through all recommended resources.

Certifications

Certification	Signal	Cost	Timeline
ISPOR Membership and Short Course Completion	High	\$125 to \$350 annual membership plus \$200 to \$800 per short course	Ongoing; complete 2 to 3 short courses in first 6 months
Graduate Certificate in HEOR or Health Outcomes Research	High	\$3,000 to \$8,000 depending on program	6 to 12 months part-time
SAS Certified Clinical Trials Programmer	Helpful	\$180 exam fee	2 to 4 months study
R Programming Certification (Johns Hopkins via Coursera or equivalent)	Helpful	\$49 per month Coursera subscription	2 to 3 months
Certified Health Data Analyst (CHDA, AHIMA)	Helpful	\$259 to \$329 exam fee	2 to 3 months study
HEOR Fellowship (Industry-sponsored, 1 to 2 years)	High	Paid position (fellow receives compensation)	1 to 2 years full-time

Learning Resources by Phase

Foundation

- [Paid] ISPOR Short Courses and eLearning: <https://www.ispor.org/conferences-education/education-training>
- [Paid] Cost-Effectiveness Analysis in Health: A Practical Approach (Peter Muennig): <https://www.amazon.com/Cost-Effectiveness-Analysis-Health-Practical-Approach/dp/1119011264>
- [Paid] UNT Health Science Center Certificate in Applied Health Outcomes Research: <https://www.unthsc.edu/college-of-pharmacy/about-us/certificate-in-applied-health-outcomes-research/>
- [Paid] Coursera Epidemiology Specialization (Johns Hopkins): <https://www.coursera.org/specializations/epidemiology>
- [Free] R for Health Data Science: https://argoshare.is.ed.ac.uk/healthyr_book/
- [Free] FDA Real-World Evidence Framework: <https://www.fda.gov/science-research/science-and-research-special-topics/real-world-evidence>

Modeling and RWE Methods

- [Paid] TreeAge Pro Software: <https://www.treeage.com/>
- [Free] NICE Decision Support Unit Technical Support Documents: <https://www.sheffield.ac.uk/nice-dsu/tsds>
- [Free] Cochrane Handbook for Systematic Reviews: <https://training.cochrane.org/handbook>

- [Free] AHRQ Real-World Evidence Resources: <https://effectivehealthcare.ahrq.gov/>
- [Paid] Value in Health Journal (ISPOR): <https://www.valueinhealthjournal.com/>
- [Paid] DistillerSR Systematic Review Software: <https://www.distillersr.com/>

Specialization

- [Paid] ISPOR Annual and European Conferences:
<https://www.ispor.org/conferences-education/conferences>
- [Free] PCORI Methodology Standards:
<https://www.pcori.org/research/about-our-research/research-methodology/pcori-methodology-standards>
- [Free] Causal Inference: What If (Hernan and Robins):
<https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>
- [Free] AMCP Format for Formulary Submissions:
<https://www.amcp.org/resource-center/format-formulary-submissions>
- [Free] Target Trial Emulation Framework Resources: <https://www.targettrialemlation.com/>

Your First Three Moves

Join ISPOR and complete your first short course on HEOR methodology (3 hours)

- Join ISPOR at the appropriate membership level (\$125 for students/early career, \$350 for professionals) and explore their eLearning catalog
- Enroll in one ISPOR short course: 'Introduction to Health Economic Modeling' or 'Systematic Reviews for HEOR' are strong starting points
- Read the FDA RWE Framework document ([fda.gov/science-research/science-and-research-special-topics/real-world-evidence](https://www.fda.gov/science-research/science-and-research-special-topics/real-world-evidence)) to understand the regulatory context driving this field

Identify your clinical-to-HEOR bridge and map target employers (2 hours)

- List 3 to 5 clinical questions from your practice that could be answered with outcomes research or real-world evidence (e.g., 'Do patients on Drug A have fewer readmissions than Drug B?'). These become your portfolio project seeds.
- Search the ISPOR Career Center (careers.ispor.org) and LinkedIn for 'HEOR Analyst' and 'RWE Analyst.' Note which companies appear most frequently and what skills they require.
- Identify whether your background maps best to Track A (modeling), Track B (RWE/observational), Track C (market access), or Track D (patient-centered outcomes). Pharmacists often start with A or C; nurses and other clinicians often start with B or D.

Start building your statistical programming skills and HEOR reading habit (Ongoing (1 to 2 hours per day))

- Begin learning R (healthyr book at argoshare.is.ed.ac.uk/healthyr_book/) or SAS (SAS Programming 1 course on SAS Academy). Complete at least one lesson per day.
- Subscribe to the Value in Health journal table of contents alerts and ISPOR newsletter. Read one HEOR paper per week, focusing on methods sections to learn how studies are designed and analyzed.

- Download and explore one public dataset relevant to your clinical area (CMS Synthetic Claims at data.cms.gov or AHRQ HCUP data) to begin understanding how real-world data is structured.

Ready to go deeper? Take the free career quiz at quiz.ehoneahobed.com or subscribe to The Transmutation newsletter at thetransmutation.substack.com